

Mounika Chanumolu
Data Engineer / Cloud Engineer

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Dallas, TX

Experienced Data Engineer with 8 years of expertise in building scalable data pipelines, optimizing data processes, and delivering actionable insights across cloud platforms. Proven ability to drive business outcomes through cloud cost efficient data solutions and collaboration with cross-functional teams.

PROFESSIONAL SUMMARY:

- Collaborated with Azure tools such as Azure Data Factory, Blob Storage, Synapse Analytics, and Databricks to process, manage, and build complex data pipelines.
- Set up and managed scalable cloud storage solutions (Azure Blob Storage, AWS S3) for handling large datasets while ensuring security and accessibility.
- Extracted data from APIs, relational databases, and unstructured file systems, transforming raw information into structured formats for processing and user accessibility.
- Designed and maintained robust ETL pipelines, ensuring accurate and consistent data delivery to support business decision-making.
- Designed serverless data pipelines using Azure Functions and AWS Lambda, enabling on-demand, scalable data processing.
- Resolved critical data challenges, such as handling missing values, handling nulls, duplicate records, inconsistencies and implementing data quality practices for reliability.
- Developed reusable pipeline templates to standardize development practices and improve project efficiency.
- Streamlined and automated data extraction, transformation, and reporting processes using Python, PySpark, and Azure Data Factory, enhancing team productivity.
- Introduced cost-optimization techniques in cloud environments by automating resource allocation and scheduling, reducing unnecessary expenditures without compromising performance.
- Monitored and maintained ETL pipelines to optimize efficiency and ensure minimal downtime for business-critical operations.
- Engineered and optimized data models for seamless cross-platform integration, enabling structured datasets for dynamic visualization and reporting.
- Constructed complex SQL queries for data extraction, aggregation, and reporting to support key business processes.
- Implemented reliable data backup and recovery protocols to safeguard critical information and ensure business continuity.
- Partnered closely with data analysts and scientists to align technical data solutions with strategic business objectives.
- Integrated third-party APIs into data solutions, enhancing datasets with enriched information for improved analytics and applications.
- Executed successful data migrations from legacy on-premise systems to modern, scalable cloud platforms, improving agility and scalability.
- Utilized Git version control systems to track and collaborate on codebase changes with distributed teams.
- Employed log monitoring tools to track ETL job execution, detect failures, and ensure uninterrupted service delivery.
- Enforced role-based access control (RBAC) and implemented security protocols to protect sensitive organizational data.
- Proactively stayed updated on emerging technologies and industry trends, continuously exploring innovations to enhance workflows and improve solution quality.

SKILLS:

- Programming & Frameworks: Python, PL/SQL, DAX, Power Query, PySpark.
- Data Engineering: Azure Data Factory, Azure Functions, Azure Blob Storage (ADLS), Azure Synapse, Azure Data Lake, Databricks, AWS Glue, AWS Lambda, AWS Athena, Azure Logic Apps, AWS SageMaker.
- Data Analysis: Data Modeling, ETL, Data Governance, Performance Optimization, Cloud Computing, Data Integration.
- Tools & Platforms: JIRA, ServiceNow, GitHub (GIT), APIs, Postman.
- Databases: MS SQL Server, Snowflake, AWS Redshift, Synapse.

EDUCATION:

- Masters in Computer Science – Kent State University, 2017
- Bachelors in Computer Science – GITAM University, 2015

PROFESSIONAL EXPERIENCE:

Walmart Global Tech Data Engineer

Feb 2023 to Present

Description:

As a Data Engineer at Walmart, I specialize in optimizing digital ad campaigns to enhance online revenue. My team is responsible for identifying and delivering the right ads to the right customers by leveraging data-driven insights to improve targeting and conversion rates.

Responsibilities:

- Utilized Azure Databricks as a powerful data processing and analytics platform for large-scale data engineering tasks.
- Performed complex data transformations, aggregations, and computations using PySpark within Azure Databricks notebooks.
- Leveraged interactive and collaborative features of Databricks notebooks for exploratory data analysis.
- Connected Azure Data Factory to various Azure data sources such as SQL, MySQL, and exported them to Azure Synapse Analytics.
- Developed and optimized data pipelines using Python and PySpark within Azure Databricks.
- Collaborated with data scientists, analysts, and stakeholders by sharing notebooks, visualizations, and insights within Databricks.
- Utilized version control systems like Git to manage and track changes to Databricks notebooks for reproducibility and collaboration.
- Created a Databricks notebook that is executed daily to pause and resume the Synapse environment, preventing unnecessary costs during non-business hours.
- Built a Databricks notebook that tracks the number of rows inserted into a Delta table daily, improving table monitoring with minimal human intervention.

Technology: Azure / Databricks / Azure SQL DB / Azure Functions / Python / PySpark / SQL / Synapse / Postman / Logic Apps / MySQL Server / Blob Storage / REST APIs.

Anthem blue cross blue shield Data Engineer

Jan 2022 to Feb 2023

Description:

Data Platform is an integrated set of technologies designed to meet an organization's end-to-end data needs—enabling seamless data acquisition, storage, preparation, delivery, and governance. I work within an Azure-based data platform focused on processing Anthem's machine-generated IT operations data.

Responsibilities:

- Developed and maintained ETL processes to ingest, process, and transform data from various sources into usable formats.
- Designed and implemented data solutions using Azure Data Factory, Azure Databricks, Azure Synapse Analytics, and other Azure data services.
- Created and optimized data models, ensuring data integrity, performance, and scalability.
- Collaborated with cross-functional teams to integrate data from multiple sources, including databases, APIs, and external data providers.
- Implemented data governance practices and ensured compliance with data security and privacy regulations.
- Monitored and optimized data pipelines for performance, efficiency, and cost-effectiveness.
- Diagnosed and resolved data-related issues, providing technical support to the offshore team.
- Maintained clear and comprehensive documentation of data pipelines, architecture, and processes.
- Stayed up-to-date with Azure Cloud technologies and recommended improvements to enhance data engineering processes.

Technology: Azure / Databricks / Azure SQL DB / Azure Functions / Python / PySpark / SQL / Synapse / Postman / Logic Apps / MySQL Server / Blob Storage / Snowflake / REST APIs.

TTEC Holdings, Inc
Azure Data Engineer

Mar 2020 to Jan 2022

Description:

TTEC's Insights Platform is a cloud-based customer data platform that provides brands with a 360° view of their customers' needs, behaviors, and preferences, offering the insights needed to deliver an exceptional customer experience.

Responsibilities:

- Developed complex pipelines using Azure Data Factory (ADF) for efficient data movement and transformation.
- Created and loaded data into Databricks Delta Lake tables for structured and scalable storage.
- Configured mount points for Azure Blob Storage (Gen1 & Gen2) to ensure seamless data access.
- Implemented password masking using Azure Key Vault to enhance data security.
- Built pipelines to move data from Azure Blob Storage to the Data Lake, ensuring data availability for analysis.
- Collaborated with Data Scientists, Business Analysts, and Partners to derive meaningful insights from data.
- Migrated data from on-premise servers to Azure Data Lake, streamlining cloud adoption.
- Utilized Python Pandas Data Frame for in-depth data analysis and processing.
- Converted SQL queries into PySpark and loaded the transformed data into Delta tables for optimized querying.
- Worked with Azure Synapse Analytics to perform advanced data analysis and reporting.

Technology: Azure / Databricks / Azure SQL DB / Azure Functions / Python / PySpark / SQL / Synapse / Postman / Logic Apps / MySQL Server / Blob Storage / REST APIs.

TransUnion
Cloud Engineer

Jun 2018 to Mar 2020

Description:

Digital authentication plays a vital role in verifying users and reducing the need for human intervention. This project aimed to read user-submitted digital data for authentication, re-verify details against the system, and ensure secure, seamless digital verification

Responsibilities:

- Designed and developed Azure-based data pipelines to support data engineering initiatives.
- Collaborated with business stakeholders to understand and implement data requirements effectively.
- Created and optimized data models to enhance analysis and reporting.
- Implemented data warehousing solutions using Azure services for efficient data storage and retrieval.
- Ensured compliance with data security and privacy standards to protect sensitive user information.
- Designed, developed, and tested software applications for authentication processes.
- Monitored and optimized data pipelines for performance, reliability, and cost-efficiency.
- Utilized PySpark for large-scale data processing and transformation.
- Developed automated scripts for error tracking and recovery mechanisms in data pipelines, ensuring smooth operations.

Technology: Azure / Databricks / Azure SQL DB / Azure Functions / Python / PySpark / SQL / Synapse / Postman / Logic Apps / MySQL Server / Blob Storage / REST APIs.

Ford Motor Company
Software Engineer

Feb 2017 to Jun 2018

Description:

This project helps in finding vehicles due for service using IoT-generated data from the vehicles. Based on this data, service campaigns are sent to respective owners, along with coupons to encourage upgrades to newer vehicle models. This approach drives sales while simplifying the process for customers to find a dealership location, creating a win-win situation for both customers and Ford.

Responsibilities:

- Built strong and lasting partnerships with business stakeholders.
- Coordinated work between onshore and offshore teams to ensure timely task completion.
- Wrote complex Hive queries to support business requirements.
- Collaborated with cross-functional teams to define, design, and implement solutions.
- Worked closely with business teams to understand their needs and translate them into technical solutions.
- Utilized Apache Hive for querying and managing data stored in Hadoop clusters, creating external tables, and writing optimized queries.
- Designed and optimized HDFS storage structures to enhance data retrieval efficiency and performance.
- Utilized cloudera for managing and monitoring Hadoop clusters, ensuring high availability and performance.
- Developed and maintained Sqoop scripts for efficient data ingestion from relational databases into Hadoop clusters.
- Implemented Python based data processing pipelines to automate data transformation and validation tasks.

Technology: *SQL / HDFS / Hive / Python / Sqoop / Cloudera.*